Why Domain Matters, part 1

Retail and grocery

The record says

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On hand: 55 bags. No reorder needed.
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What actually happened

The shelf has been empty since Wednesday.

Thursday 11 June, 7:40 pm. Store 014, Jayanagar.

The 5 kg atta bay has been empty since 3:48 on Wednesday afternoon.

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Shelf walk, Thu 11 Jun 19:40
Atta 5 kg bay: empty
Handheld on hand: 55
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A customer asks. Staff check the backroom, find nothing, and say it must be on the next truck. No truck is coming: the reorder system thinks the store has 55 bags.

Five readings of one shelf

Who	What they see	What they conclude
Store manager	An empty bay, and a handheld that says 55	“It must be on the next truck.”
Reorder system	55 bags on the books, reorder point 40	“In stock. No reorder needed.”
Demand planner	Sales down 59% in a week; forecast 24.3 → 20.2 a day	“Demand is falling. Trim the order.”
Category manager	5 kg share of atta units: 40% → 25%	“Shoppers are moving to 1 kg.”
Supplier’s sales rep	Store orders: 172 → 112 bags a week	“A competitor is taking the space.”

Each conclusion is reasonable given the record in front of that person. Each is wrong.

What actually happened

Tue 9 Jun, 06:17

14 cases, 56 bags, arrive. The books go up by 56. The cases go behind the rice, in the wrong bay.

Wed 10 Jun, 15:48

The shelf runs out. The books still show 56 bags.

Wed to Fri, 22:00

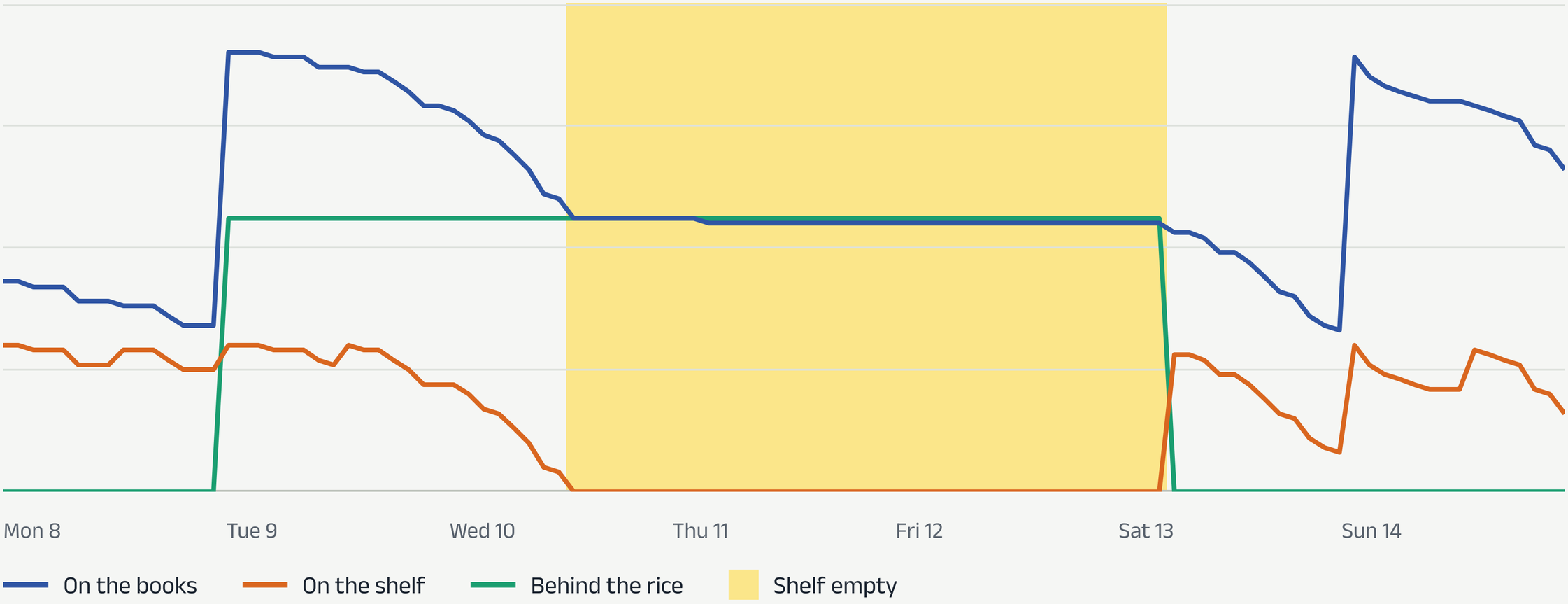
Each night the reorder run sees 55 or 56 bags against a reorder point of 39 or 40, and orders nothing.

Sat 13 Jun, 10:00

Staff restacking the rice find the cases.

39 open hours with an empty shelf, and stock in the building the whole time.

The books never fell to the reorder point



5 kg atta at Store 014, bags, every open hour from 8 to 14 June 2026. Grid lines every 25 bags. Synthetic world, seed 42.

Demand never fell. The shelf did.

157

bags customers wanted in the week of 8 June

88

bags the tills recorded

71 bags wanted during the gap and not bought

What the shopper did	Bags
Bought another brand	26
Went to another store	19
Took a different pack size	13
Came back later	8
Never bought it	5

True demand averaged 24.7 bags a day in May and 24.9 in June. Shopper responses follow Gruen, Corsten and Bharadwaj (2002).

This is the normal case

8.3%

of grocery items were out of stock on average, worldwide

72%

of out-of-stocks started in the store: 47% ordering and forecasting, 25% shelf management

65%

of 370,000 inventory records at 37 stores of one retailer were wrong

The world behind this deck matches: shelves empty 5.0% of open hours, 74% of those hours starting in the store, and 8 of 11 records wrong at the month-end count.

Sources: FMI, GMA and CIES worldwide out-of-stock study (2002); DeHoratius and Raman (2008), Management Science 54(4).

One name, four numbers

97%

The system showed stock

Open hours with stock on the books

75%

Shelf audits found it

8 audits, Tuesdays and Fridays at 11:00

77%

It was on the shelf

Open hours with a bag on the shelf

78%

Customers found it

Bags wanted that were on the shelf

On-shelf availability of 5 kg atta at Store 014 in June. None of these is wrong. A dashboard that shows one without saying which one it is, is.

The forecast learns the wrong lesson

Learning from recorded sales

24.3 → 20.2

bags a day, from before the gap to the end of June

Error against recorded sales: 54%

Filling in the empty-shelf hours

25.7

bags a day at the end of June

Error against what customers wanted: 33%

Recorded sales are capped by what was on the shelf. The planner sees a volatile item; the truth is a steady item with a broken shelf.

Same store, different item: milk

Delivered by the dairy at 05:30 and good for 48 hours from packing. No reorder point to freeze; the risk runs the other way.

Toned milk standing order	Thrown away	Wanted, not found	Margin after waste
15% less	0	1,279	₹9,996
10% less	0	750	₹10,586
5% less: the most margin	13	269	₹10,810
As today	291	119	₹4,121
5% more	668	70	−₹5,132
10% more	1,033	27	−₹14,100
15% more	1,432	0	−₹23,925

Pouches in June, same customers in every row. Assumes ₹1.30 earned per pouch sold and ₹24.70 lost per pouch thrown away.

Detection helps. Prevention helps more.

A low-sales alert, checked next morning

157 → 90

bags customers could not buy in June. Empty open hours fall from 97 to 59.

It still waits a day, and one mis-scanned pack can hide an empty shelf from it.

Scan every case into a backroom bin

157 → 1

bag customers could not buy in June. Nothing goes missing behind the rice.

The put-away scan also catches the 22 June pallet labelled 5 kg that held 1 kg packs.

Same customers in every scenario; the changes start on 1 June. Try both in the demo.

Nothing in the records says “misplaced”

Night	On hand	Reorder	Order
9 Jun	78	39	0
10 Jun	56	39	0
11 Jun	55	40	0
12 Jun	55	40	0
13 Jun	33	40	56

The signal is a pattern across tables: book stock stops moving when sales stop, with no delivery, write-off or count in between.

Each table is consistent on its own. The story only appears when you join them, and knowing which join to make comes from knowing how a store runs.

The reorder run at 22:00, from Store 014’s forecast table. Identifiers in the data use GS1 prefix 29, set aside for restricted circulation.

How to learn this domain

How stores run	Raman, DeHoratius and Ton, *Execution: The Missing Link in Retail Operations* (2001). Then walk a supermarket and note where stock lives.
Inventory policy	Reorder points, min/max, safety stock, service levels. Then stock loss stalling replenishment: Yun Kang's MIT thesis (2004).
Forecasting	Hyndman and Athanasopoulos, *Forecasting: Principles and Practice* (free online), then the M5 Walmart data on Kaggle.
Store data	Till lines, book stock, receipts, counts and audits, joined by GTIN, GLN and SSCC.
Practice	Download Store 014's June records and try the four exercises before opening the answer key.

A rough plan for a few evenings a week over about two months. Links in the post: kvinayakpai.github.io/domain-series/retail

Every system keeps a record. Domain knowledge is knowing when the record is wrong.

Run the scenarios and download the records:

kvinayakpai.github.io/domain-series

> **Next: logistics**
>
> The truck was on time. The pallet was wrong.